Complementary-Filtered Residual World Models for Receding-Horizon Control of Nonlinear Systems

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***Abstract*—Receding-horizon control with a learned dynamics model is sensitive to both forecast-origin error and transition error. This paper presents a Complementary-Filtered Residual World Model (CFR-WM) that uses a mask-gated fixed-gain context update before free rollout with nominal dynamics and a bounded neural residual. On five nonlinear controlled systems, three test regimes, and eight paired seeds, CFR-WM reduces 25-step normalized root mean squared error by 8.97% relative to a physics-guided neural baseline (95% interval [6.88%, 10.95%], $p = 0.00391$ against a 5% margin). The corrupt-regime improvement is 12.05%, while the clean-regime noninferiority criterion is not met. In 1920 paired receding-horizon control episodes, CFR-WM reduces realized normalized state loss by 1.25% (95% interval [0.89%, 1.59%], $p = 0.00391$) relative to the same baseline. The filter-plus-nominal component slightly outperforms the complete model. Measured NanoDrone flights show lower 50-step position error than learned comparators but higher full-state error than the matched unfiltered model and undercoverage of predictive intervals. The evidence identifies context filtering, rather than the learned residual or uncertainty output, as the main source of robustness under corrupted observations.**

NOTE TO PRACTITIONERS

This work addresses automation systems that optimize actions from short, imperfect state histories while retaining a usable nominal process model. The proposed interface first reconciles recent measurements with the nominal model, then supplies the corrected state to a predictive controller. It can be inserted between an existing state stream and a trajectory optimizer without training a policy. Across the tested nonlinear plants, this update is most useful when observations are noisy or missing; it does not improve every clean operating condition. A practical deployment should therefore validate the filter gain for each sensor regime and retain the nominal rollout as a required ablation. In our experiments, the neural residual did not improve mean point accuracy over filtering plus nominal dynamics, and the reported predictive intervals were too narrow on measured flights. The current uncertainty output should not be used directly for safety margins. Before deployment, operators should test longer sensor outages, plant-specific constraints, computation time on the target controller, and closed-loop behavior on the physical system.

***Index Terms*—Complementary filtering, model predictive control, nonlinear systems, physics-guided learning, state estimation, world models.**

I. INTRODUCTION

Model predictive control (MPC) repeatedly queries a dynamics model while optimizing an action sequence. Its performance therefore depends on two coupled quantities: the state at the forecast origin and the transition used to propagate that state under candidate controls [1]. For a learned transition, improving one quantity does not guarantee improving the other. A model may forecast accurately from clean states yet initialize poorly from incomplete measurements, or it may correct the current state while accumulating transition error during rollout.

World models provide trainable predictors for planning and continuous control [2], [3]. Physics-guided models add a nominal transition or physical constraint, which can reduce data demand and expose departures from the assumed mechanism [4], [5]. Neither construction by itself resolves how recent noisy or missing observations should be reconciled with nominal dynamics before planning.

This paper studies that interface through the Complementary-Filtered Residual World Model (CFR-WM). During an eight-step observed context, nominal dynamics predict the next state and a mask-gated complementary update combines the prediction with available measurements. The final corrected state initializes a probabilistic nominal-physics-plus-neural-residual transition. Future states are then rolled out freely under a candidate control sequence. The design separates forecast-origin correction from transition learning and can be queried directly by an optimizer.

The evaluation connects predictive accuracy to realized regulation. The open-loop study contains five nonlinear controlled systems, three data regimes, eight independently generated data/training seeds, 11 model variants, and 440 trained models. A frozen additional study places four prediction interfaces inside the same cross-entropy MPC controller and evaluates 1920 paired control episodes. An external experiment uses measured NanoDrone states [6] with a deliberately reduced quadrotor prior to test transfer beyond the synthetic equations.

The contributions are:

1) a causal estimator–predictor architecture that combines a mask-gated complementary context update, nominal dynamics, and a bounded neural residual for free multistep rollout;
2) an error decomposition and constrained mean-square-optimal scalar gain that identify when a fixed complementary update should help;
3) a seed-paired evaluation that links open-loop forecast error to receding-horizon regulation without retraining or controller-specific model selection; and
4) component, outage, calibration, and measured-flight analyses that show where the context filter helps and where the residual and predictive intervals do not.

The results do not reduce to a single accuracy claim. The aggregate open-loop contrast favors CFR-WM, whereas the clean-regime gate fails; the fixed gain-0.5 filter followed by nominal rollout outperforms the learned residual on aggregate point error; and the measured-flight study separates position error from full-state error and interval coverage. These distinctions determine how the model should be used in an automation pipeline.

II. Related Work

A. State estimation and learned dynamics

The Kalman filter balances a dynamic prediction against a measurement through an error-covariance-dependent gain [7]. Complementary filters use a simpler structured allocation between the two information sources and can be effective when their error characteristics differ by channel or frequency [8]. Neural estimators such as KalmanNet learn parts of this recursion for partially known dynamics [9]; differentiable ensemble filters and the broader combination of machine learning with data assimilation provide related routes to trainable state estimation [10], [11]. CFR-WM retains an explicit fixed-gain update: the gain is interpretable, missing channels receive no direct measurement correction, and the transition model remains independently testable.

Physics-guided learning combines measured data with mechanistic structure through features, constraints, differentiable solvers, or residual dynamics [12], [13]. Hybrid first-principles/data-driven modeling has been studied as a general design pattern and in nonlinear dynamics and online vehicle modeling [14], [15], [16]. Residual models are attractive for control because the nominal transition can remain operational outside the training distribution. Their learned correction can nevertheless amplify error when model mismatch changes. We therefore evaluate the filter-plus-nominal component and vary the residual multiplier at fixed trained weights.

B. World models and predictive control

Latent and recurrent world models have achieved strong performance on broad control suites, including planning with learned representations [17], [2]. Dynamics-model learning has also been surveyed for robotic manipulation and used in predictive autonomous-driving control [18], [19]. These systems commonly learn task rewards, latent dynamics, or policies jointly. The present study instead uses observed physical states, known controls, and a separate MPC objective so that forecast-origin estimation and transition error can be examined directly.

Learning-based MPC requires a model that remains useful over optimized action sequences, not only under the data-collection policy [1]; safety and real-world operating constraints remain central in learning-based control [20], [21]. Sparse system identification has likewise been connected to MPC in low-data nonlinear settings [22]. Our closed-loop experiment holds the optimizer, horizon, action constraints, objective, and random draws fixed across prediction interfaces. It therefore tests the downstream value of the model change rather than a jointly tuned controller-model package.

III. Complementary-Filtered Residual World Model

A. Problem formulation

Let $\mathbf{x}_t \in \mathbb{R}^{d_x}$ be a physical state and $\mathbf{u}_t \in \mathbb{R}^{d_u}$ a control. The available observation is

$$\mathbf{y}_t = \mathbf{m}_t \odot (\mathbf{x}_t + \boldsymbol{\epsilon}_t), \tag{1}$$

where $\mathbf{m}_t \in \{0,1\}^{d_x}$ is a channel mask. Missing entries are filled by the latest available value, giving $\tilde{\mathbf{y}}_t$; the mask and elapsed time since observation $\boldsymbol{\delta}_t$ remain model inputs. Given an eight-step context and a candidate future control sequence, the model predicts H future states without teacher forcing.

B. Mask-gated context update

Let F_0 be a nominal dynamics step. Starting from the first imputed context state, the filter alternates nominal prediction and measurement correction:

$$\bar{\mathbf{x}}_k = F_0(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \tag{2}$$

$$\mathbf{K}_k = \alpha \mathbf{m}_k, \qquad \alpha = 0.5, \tag{3}$$

$$\hat{\mathbf{x}}_k = \Pi[\bar{\mathbf{x}}_k + \mathbf{K}_k \odot (\tilde{\mathbf{y}}_k - \bar{\mathbf{x}}_k)]\,. \tag{4}$$

An observed channel receives equal pre-projection weight from the nominal prediction and measurement. A missing channel receives zero direct correction. Π is the identity except for the pendulum sine–cosine representation, which is projected to the unit circle. The gain is fixed independently of validation and test outcomes.

A gated recurrent unit (GRU) simultaneously encodes $[\tilde{\mathbf{y}}_k, \mathbf{m}_k, \log(1+\boldsymbol{\delta}_k), \mathbf{u}_{k-1}]$. Its final hidden state carries trajectory context into the rollout, but it does not alter the measurement gain.

C. Nominal-plus-residual rollout

At future step h, the transition computes

$$\boldsymbol{\mu}^p_{t+h} = F_0(\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}), \tag{5}$$

$$[\mathbf{r}_{t+h}, \ell_{t+h}] = R_\theta([\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}, \mathbf{b}_{t+h-1}]), \tag{6}$$

$$\boldsymbol{\mu}_{t+h} = \Pi\big[\boldsymbol{\mu}^p_{t+h} + 0.25\tanh(\mathbf{r}_{t+h})\big]\,, \tag{7}$$

$$\boldsymbol{\sigma}^2_{t+h} = \max\{\exp(\mathrm{clip}(\ell_{t+h}, -8, 4)), 10^{-6}\}. \tag{8}$$

The residual is bounded to one quarter of a training standard deviation per step in normalized coordinates. Predicted means feed the next transition, and a GRUCell updates $\mathbf{b}$. The model minimizes mean diagonal Gaussian negative log likelihood over all states and all 25 rollout steps:

$$\mathcal{L} = \frac{1}{2Hd_x}\sum_{h=1}^{H}\sum_{q=1}^{d_x}\left[\log(2\pi\sigma^2_{hq}) + \frac{(x_{hq}-\mu_{hq})^2}{\sigma^2_{hq}}\right]. \tag{9}$$

D. Gain and error propagation

For one observed scalar channel before a nonidentity projection, let $e_p = \bar{x} - x$ be nominal prediction error and $\epsilon = y - x$ measurement error. The filtered error is $e_f = (1-\alpha)e_p + \alpha\epsilon$. With $v_p = \mathbb{E}[e_p^2]$, $v_\epsilon = \mathbb{E}[\epsilon^2]$, and $c = \mathbb{E}[e_p\epsilon]$,

$$\mathbb{E}[e_f^2] = (1-\alpha)^2 v_p + \alpha^2 v_\epsilon + 2\alpha(1-\alpha)c. \tag{10}$$

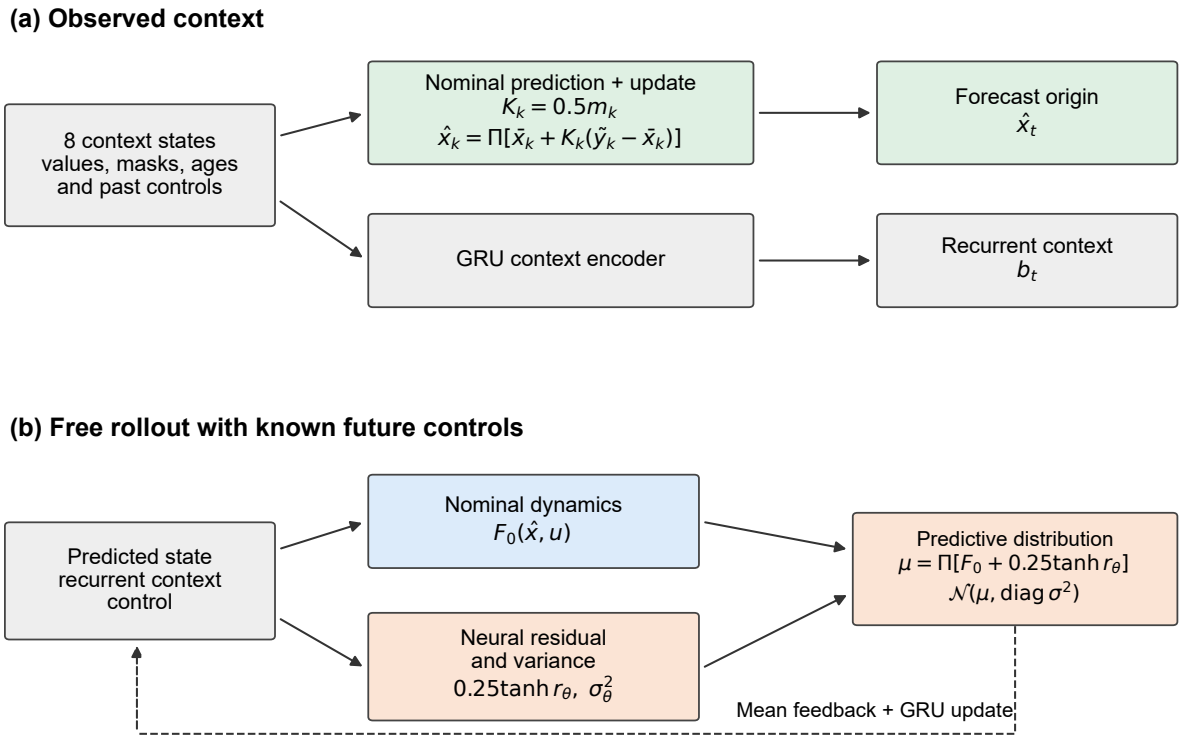


Fig. 1. Causal estimator–predictor interface. Nominal propagation and a mask-gated complementary update process the observed context. The corrected forecast origin initializes a nominal-plus-residual free rollout under a candidate future control sequence. No future observation is used during a rollout.

If $v_p + v_\epsilon - 2c > 0$, its constrained minimizer is

$$\alpha^* = \mathrm{clip}_{[0,1]}\left(\frac{v_p - c}{v_p + v_\epsilon - 2c}\right). \tag{11}$$

Thus $\alpha = 0.5$ is optimal only when $v_p = v_\epsilon$ and $c = 0$. This identity predicts regime dependence: cleaner measurements favor a larger gain, whereas inaccurate nominal propagation favors stronger measurement correction.

If the rollout transition is locally L-Lipschitz with local error η_h,

$$\|\hat{\mathbf{x}}_{t+H} - \mathbf{x}_{t+H}\| \le L^H \|\hat{\mathbf{x}}_t - \mathbf{x}_t\| + \sum_{h=0}^{H-1} L^{H-1-h}\eta_h. \tag{12}$$

The filter changes the initial-error term; the residual changes the accumulated transition term. The component experiments estimate their separate effects.

E. Receding-horizon use

For each control step, the current eight-state context is processed once per candidate sequence. Cross-entropy MPC samples bounded action sequences, evaluates their predicted mean trajectories, retains the lowest-cost elites, and applies the first action of the updated mean. The next optimization is warm started by shifting that mean. The model is not retrained for the control objective, and predictive variances are not used by this controller.

IV. Evaluation Design

A. Open-loop controlled benchmark

The benchmark contains a damped oscillator, forced Duffing oscillator, controlled Van der Pol oscillator, torque-driven pendulum, and cart–pole. Fourth-order Runge–Kutta integration generates training, validation, and test trajectories. Each of eight seeds independently draws the data and initializes training. Methods share each seed–system data set.

Three test regimes are fixed: clean in-distribution (ID), corrupted ID, and corrupted out-of-distribution (OOD) parameters. Corruption combines additive measurement noise and channel missingness with causal carry-forward. Eleven variants include recurrent, transformer, neural ODE, physics-guided, nominal, and residual predictors plus filter-gain and training-horizon sensitivities. Every learned method receives the same six-trial validation budget. The primary metric is normalized root mean squared error at the 25-step horizon (NRMSE-25), averaged equally across the 15 system–regime cells within each seed.

The predeclared primary contrast compares CFR-WM with the PGNN-style model. It requires a paired 95% interval lower bound above a 5% improvement and an exact one-sided seed sign-flip $p < 0.05$. The gated robustness claim additionally requires improvement in the two corrupted regimes and noninferiority within 5% on clean ID data. Component analyses compare the fixed filter plus nominal rollout with the complete model, vary the residual multiplier at fixed weights, and impose a synchronous two-step context outage.

B. Frozen closed-loop regulation study

The additional control study was specified before execution and reuses the frozen PGNN-style and CFR-WM checkpoints. It includes all five systems, eight seeds, and three plant/observation conditions: ID clean, ID corrupt, and OOD-parameter corrupt. Four independently generated plant episodes populate each seed–system–condition cell. Initial states, plant parameters, observation draws, masks, and planner draws are paired across four prediction interfaces: nominal propagation from the last imputed state, filter plus nominal propagation, PGNN-style, and CFR-WM. ID clean and ID corrupt also share the plant draw.

Seven zero-input transitions establish the eight-state context. The controller then regulates for 20 steps. Deterministic-mean cross-entropy MPC uses horizon 8, 64 candidates, two iterations, and eight elites. Action sequences start with standard deviation 0.6 of the action limit, with a 0.05 floor. The stage objective is mean squared state error in training-standard-deviation units plus 0.01 times squared normalized action; terminal state error receives an additional factor of two. The pendulum target is $(\sin\theta, \cos\theta, \omega) = (0, 1, 0)$ and the other targets are zero. Corrupt observations use Gaussian noise with scalar standard deviation 8% of the mean training-state standard deviation and 25% independent channel missingness.

The primary closed-loop response is the realized mean normalized state loss over 20 controlled transitions. Episodes, systems, and conditions receive equal weight within each seed. The reported CFR-WM–PGNN contrast is the mean of eight seed-level relative improvements, with a 5,000-replicate paired seed bootstrap interval and an exact one-sided sign-flip test against zero. Controller settings were not changed after observing outcomes.

C. Measured-flight external evaluation

The NanoDrone experiment uses recorded states and motor commands from the nonlinear system-identification benchmark [6]. Three held-out flights from a distinct trajectory family provide 387 forecast windows. A reduced quadrotor prior omits parts of the recorded platform dynamics; learned models have 47,000 parameters and use five training seeds. The evaluation crosses four context conditions (recorded, added noise, missingness, and both), two horizons (25 and 50), and four predictors (CFR-WM, its unfiltered residual counterpart, a recurrent model, and the nominal prior). It reports full-state NRMSE, position RMSE, velocity RMSE, and marginal 95% coverage. Flights receive equal weight before averaging training seeds.

Training, test, and analysis artifacts are hash-checked before tables and figures are generated. The supplementary material gives the system equations, parameter ranges, tuning space, complete cell results, and artifact inventory.

V. Results

A. Open-loop prediction

Table I summarizes the cell-balanced confirmatory results. At 25 steps, CFR-WM obtains NRMSE 0.0998 versus 0.1096 for the preselected PGNN-style comparator. The paired seed-level relative improvement is 8.97% (95% fixed-cell bootstrap interval [6.88%, 10.95%]); the exact one-sided sign-flip test against the 5% practical margin gives $p = 0.00391$. The corresponding absolute difference is -0.0098 (95% interval [-0.0121, -0.0075]). These quantities satisfy the predeclared aggregate criterion.

The regime split is asymmetric. Combining the corrupt-ID and corrupt-OOD cells gives a 12.05% improvement (95% interval [10.51%, 13.49%], $p = 0.00391$ against 5%). On clean ID data the change is -13.08% (95% interval [-20.60%, -5.67%], $p = 0.95703$ against the -5% noninferiority margin). The clean component therefore does not satisfy the predeclared gated robustness criterion. The system-level contrasts are also heterogeneous: three clean-system cells deteriorate by more than 10%, whereas all five corrupt-ID cells favor CFR-WM.

B. Component and outage analyses

Table II separates forecast-origin correction from residual rollout. Filtering followed by nominal propagation reaches aggregate NRMSE 0.111, compared with 0.119 for CFR-WM and 0.132 for PGNN-style. Loading the trained CFR-WM checkpoints and setting the inference residual multiplier to zero reproduces the filter-plus-nominal result; increasing the multiplier from 0 to 0.5 increases mean error in every reported regime. Thus the learned residual does not improve point accuracy in this experiment.

A synchronous two-step all-channel context outage increases CFR-WM NRMSE by 2.7%, from its matched clean reference, whereas PGNN-style increases by 51.9%. This intervention isolates missing context observations; it does not test longer outages or actuator faults.

C. Receding-horizon regulation

Table III reports realized control outcomes over 1920 paired episodes. CFR-WM yields state loss 0.3703 versus 0.3750 for PGNN-style. The seed-balanced relative improvement is 1.25% (95% paired bootstrap interval [0.89%, 1.59%]); the exact one-sided sign-flip test against zero gives $p = 0.00391$. Mean improvements favor CFR-WM in all 15 system–condition cells, ranging from 0.006% for the damped-oscillator OOD-corrupt cell to 2.8% for the cart–pole corrupt-ID cell (Fig. 2).

The absolute separation between learned predictors is small. Filter plus nominal dynamics obtains state loss 0.3698, slightly below CFR-WM's 0.3703, and also uses less action effort (0.398 versus 0.404). The nominal-last interface reaches 0.3718. Closed-loop performance therefore supports the value of context correction, but not an additional benefit from the learned residual.

D. Measured-flight transfer

On held-out NanoDrone flights, CFR-WM improves 50-step position RMSE over the matched unfiltered residual model in all four context conditions (Table IV). For example, under combined noise and missingness, position RMSE is $0.094 \pm$

TABLE I
OPEN-LOOP PREDICTION OVER FIVE SYSTEMS, THREE REGIMES, AND EIGHT PAIRED SEEDS. ENTRIES ARE MEAN ± SD ACROSS SEED-LEVEL CELL-BALANCED AGGREGATES.

Method	Mean parameters	NRMSE-1	NRMSE-10	NRMSE-25
CFR-WM	57694	0.039 ± 0.001	0.057 ± 0.002	**0.100 ± 0.004**
CFR-gain-0.25	57694	0.037 ± 0.001	0.059 ± 0.002	0.103 ± 0.004
CFR-gain-0.75	57694	0.047 ± 0.001	0.065 ± 0.002	0.107 ± 0.005
PGNN-style	61275	0.060 ± 0.001	0.070 ± 0.002	0.110 ± 0.003
CFR-gain-1	57694	0.059 ± 0.001	0.073 ± 0.002	0.114 ± 0.004
PG-DMM-style	61493	0.063 ± 0.001	0.075 ± 0.002	0.114 ± 0.003
RSSM-style	62485	0.073 ± 0.002	0.090 ± 0.005	0.123 ± 0.006
Transformer-style	59708	0.074 ± 0.002	0.094 ± 0.004	0.132 ± 0.009
SimNorm-style	62485	0.074 ± 0.002	0.098 ± 0.006	0.138 ± 0.013
CFR-gain-0	57694	0.081 ± 0.003	0.100 ± 0.003	0.144 ± 0.005
CFR-one-step	57694	0.040 ± 0.001	0.136 ± 0.050	0.278 ± 0.104

TABLE II
CLASSICAL COMPONENTS AND RESIDUAL-SCALE SENSITIVITY. THE LOWER BLOCK CHANGES ONLY THE INFERENCE-TIME RESIDUAL SCALE OF TRAINED CFR-WM CHECKPOINTS.

Analysis and variant	All	Clean	ID corrupt	OOD corrupt
Nominal, final imputation	0.137	0.040	0.151	0.221
Complementary filter + nominal	0.111	0.042	0.092	0.201
PGNN-style	0.132	0.052	0.134	0.211
CFR-WM	0.119	0.055	0.097	0.204
CFR-WM, inference scale 0	0.111	0.042	0.092	0.201
CFR-WM, inference scale 0.125	0.113	0.046	0.093	0.202
CFR-WM, inference scale 0.25	0.119	0.055	0.097	0.204
CFR-WM, inference scale 0.5	0.135	0.079	0.113	0.213

TABLE III
CLOSED-LOOP REGULATION WITH A COMMON CROSS-ENTROPY MPC CONTROLLER. STATE LOSS AND ACTION EFFORT ARE MEAN ± SD ACROSS EIGHT SEED-LEVEL AGGREGATES. CONDITION COLUMNS REPORT STATE LOSS.

Controller	State loss	Action effort	ID clean	ID corrupt	OOD corrupt
Nominal, last state	0.372 ± 0.089	0.448 ± 0.013	0.362 ± 0.099	0.368 ± 0.098	0.385 ± 0.093
Filter + nominal	0.370 ± 0.090	0.398 ± 0.013	0.362 ± 0.099	0.364 ± 0.099	0.383 ± 0.092
PGNN-style	0.375 ± 0.091	0.464 ± 0.017	0.367 ± 0.100	0.370 ± 0.099	0.388 ± 0.094
CFR-WM	0.370 ± 0.090	0.404 ± 0.016	0.363 ± 0.099	0.364 ± 0.099	0.384 ± 0.092

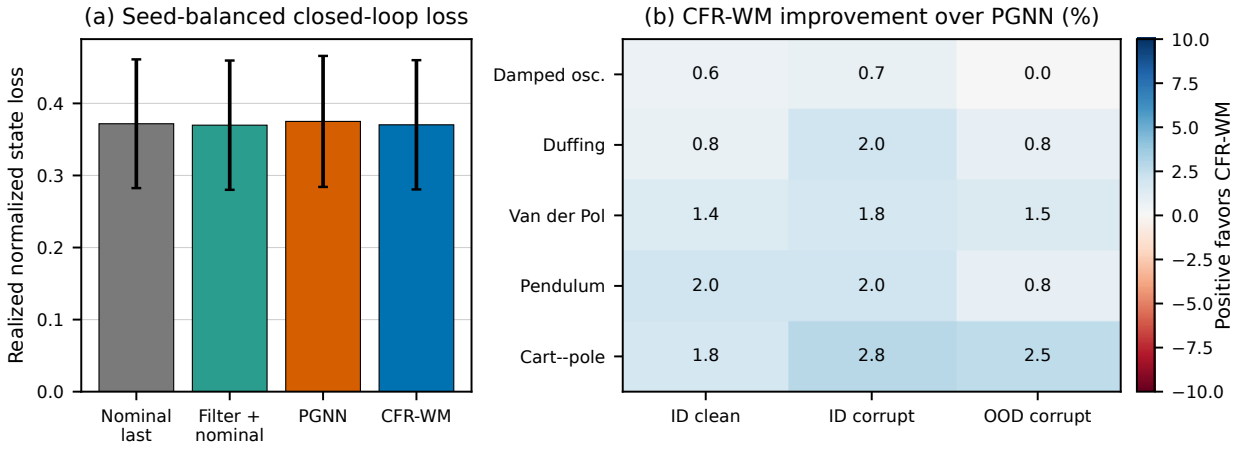


Fig. 2. Frozen closed-loop study. (a) Seed-balanced mean normalized state loss; error bars are one SD across eight seeds. (b) Mean CFR-WM improvement over PGNN-style within each system–condition cell. Positive values favor CFR-WM.

0.004 m versus 0.125 ± 0.017 m. The state components do not move uniformly: the unfiltered model has lower 50-step full-state NRMSE in all eight context–horizon comparisons, and it has lower velocity error for recorded and missing-only contexts.

The probabilistic output is not calibrated for this transfer.

TABLE IV
EXTERNAL 50-STEP PHYSICAL ERRORS ON THREE HELD-OUT NANODRONE FLIGHTS. ENTRIES ARE MEAN ± SD ACROSS FIVE TRAINING SEEDS AFTER EQUAL WEIGHTING OF FLIGHTS.

	Position RMSE (m)		Velocity RMSE (m/s)	
Context	CFR-WM	Unfiltered	CFR-WM	Unfiltered
As recorded	0.078 ± 0.003	0.083 ± 0.010	0.249 ± 0.015	0.233 ± 0.009
Noise	0.091 ± 0.003	0.115 ± 0.007	0.257 ± 0.014	0.258 ± 0.007
Missing	0.081 ± 0.005	0.095 ± 0.022	0.259 ± 0.014	0.243 ± 0.009
Noise + missing	0.094 ± 0.004	0.125 ± 0.017	0.267 ± 0.014	0.268 ± 0.009

CFR-WM's nominal 90% marginal coverage ranges from 0.524 to 0.667 across the four context conditions, compared with 0.565–0.779 for the matched unfiltered model. The same CFR-WM intervals are narrower and have higher calibration error in every condition. These intervals are therefore not used in the MPC objective.

VI. DISCUSSION

The experiments distinguish two mechanisms that are often combined inside a single learned dynamics model. Complementary context correction reduces the forecast-origin error in (12); the learned residual is intended to reduce accumulated transition error. Here the first mechanism is consistently useful under observation corruption. It explains the 12.05% open-loop gain, the outage response, and the reduction in closed-loop action effort relative to PGNN-style. Its small closed-loop state loss advantage is repeated across all system–condition means, although the aggregate magnitude is only 1.25%.

The residual evidence points in the other direction. Filter plus nominal dynamics is better than the full model in both the post-hoc open-loop analysis and the frozen closed-loop study. This result is compatible with a nominal model that is accurate enough that the learned correction adds estimation variance or compounds small one-step biases. Residual learning may still be useful when mismatch is larger, but the current benchmark does not demonstrate that benefit. A model-selection rule for automation should consequently retain the zero-residual variant rather than assuming that an additional trainable component improves control.

Equation (11) also explains the clean-regime result. A fixed gain of 0.5 assumes a particular relationship between nominal-prediction and measurement errors. When measurements are clean, equal weighting can discard useful measurement information; when measurements are corrupted, it can suppress noise and bridge missing channels. A channel- and state-dependent gain estimated without access to test outcomes is a direct extension. Its evaluation should retain the same clean noninferiority gate because an adaptive estimator can otherwise trade nominal accuracy for corruption robustness.

The measured-flight study narrows the transfer claim. Position error improves, whereas full-state error and interval coverage do not. The reduced quadrotor prior omits platform effects, and the diagonal Gaussian head does not represent cross-state or temporal dependence. Risk-sensitive control therefore requires recalibration or a richer predictive distribution before using these uncertainties as constraint margins.

The closed-loop experiment uses simulated plants, one deterministic-mean CEM configuration, 20 control steps, and a quadratic regulation objective. The measured flights provide real sensor and actuation records but not an online intervention. Physical closed-loop experiments, task constraints, longer missing-data bursts, informative sensor failures, and latency on the target controller remain necessary to assess deployment performance. The bounded residual and finite-horizon results do not by themselves establish closed-loop stability.

VII. CONCLUSION

CFR-WM separates a mask-gated complementary state update from a nominal-plus-residual free rollout. Across a controlled open-loop benchmark, the interface improves 25-step prediction under corrupted observations but does not meet the clean-regime noninferiority criterion. In a frozen receding-horizon study it reduces state loss relative to PGNN-style by 1.25%, while filter plus nominal dynamics remains slightly better than the complete model. Measured flights show a position benefit alongside worse full-state error and undercovered intervals. For the tested automation setting, complementary context filtering is the supported design element; residual and uncertainty components require stronger evidence before they are used for control decisions.

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REFERENCES

[1] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger, "Learning-based model predictive control: Toward safe learning in control," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 269–296, 2020. [Online]. Available: https://doi.org/10.1146/annurev-control-090419-075625

[2] N. Hansen, H. Su, and X. Wang, "TD-MPC2: Scalable, Robust World Models for Continuous Control," *arXiv preprint*, 2023. [Online]. Available: https://arxiv.org/abs/2310.16828

[3] D. Hafner, J. Pašukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, 2025. [Online]. Available: https://doi.org/10.1038/s41586-025-08744-2

[4] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, 2021. [Online]. Available: https://doi.org/10.1038/s42254-021-00314-5

[5] R. Yu and R. Wang, "Learning dynamical systems from data: An introduction to physics-guided deep learning," *Proceedings of the National Academy of Sciences*, 2024. [Online]. Available: https://doi.org/10.1073/pnas.2311808121

[6] R. Busetto, E. Cereda, M. Forgione, G. Maroni, D. Piga, and D. Palossi, "Nonlinear system identification for a nano-drone benchmark," *Control Engineering Practice*, vol. 172, p. 106871, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0967066126001152

[7] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960. [Online]. Available: https://doi.org/10.1115/1.3662552

[8] R. Mahony, T. Hamel, and J.-M. Pflimlin, "Nonlinear complementary filters on the special orthogonal group," *IEEE Transactions on Automatic Control*, vol. 53, no. 5, pp. 1203–1218, 2008. [Online]. Available: https://doi.org/10.1109/TAC.2008.923738

[9] G. Revach, N. Shlezinger, X. Ni, A. L. Escoriza, R. J. G. van Sloun, and Y. C. Eldar, "KalmanNet: Neural Network Aided Kalman Filtering for Partially Known Dynamics," *IEEE Transactions on Signal Processing*, 2022. [Online]. Available: https://doi.org/10.1109/TSP.2022.3158588

[10] Y. Chen, D. Sanz-Alonso, and R. Willett, "Autodifferentiable ensemble kalman filters," *SIAM Journal on Mathematics of Data Science*, vol. 4, no. 2, pp. 801–833, 2022. [Online]. Available: https://doi.org/10.1137/21M1434477

[11] S. Cheng, C. Quilodrán-Casas, S. Ouala, A. Farchi, C. Liu, P. Tandeo, R. Fablet, D. Lucor, B. Iooss, J. Brajard, D. Xiao, T. Janjić, W. Ding, Y. Guo, A. Carrassi, M. Bocquet, and R. Arcucci, "Machine Learning With Data Assimilation and Uncertainty Quantification for Dynamical Systems: A Review," *IEEE/CAA Journal of Automatica Sinica*, 2023. [Online]. Available: https://doi.org/10.1109/jas.2023.123537

[12] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems," *ACM Computing Surveys*, 2022. [Online]. Available: https://doi.org/10.1145/3514228

[13] W. Liu, Z. Lai, K. Bacsa, and E. Chatzi, "Physics-guided Deep Markov Models for learning nonlinear dynamical systems with uncertainty," *Mechanical Systems and Signal Processing*, 2022. [Online]. Available: https://doi.org/10.1016/j.ymssp.2022.109276

[14] W. A. Bradley, J. Kim, Z. Kilwein, L. Blakely, M. Eydenberg, J. Jalvin, C. D. Laird, and F. Boukouvala, "Perspectives on the integration between first-principles and data-driven modeling," *Computers & Chemical Engineering*, 2022. [Online]. Available: https://doi.org/10.1016/j.compchemeng.2022.107898

[15] H. Robinson, S. Pawar, A. Rasheed, and O. San, "Physics guided neural networks for modelling of non-linear dynamics," *Neural Networks*, 2022. [Online]. Available: https://doi.org/10.1016/j.neunet.2022.07.023

[16] H. Chen, S. Lou, and C. Lv, "Hybrid physics-data-driven online modelling: Framework, methodology and application to electric vehicles," *Mechanical Systems and Signal Processing*, 2022. [Online]. Available: https://doi.org/10.1016/j.ymssp.2022.109791

[17] T. M. Moerland, J. Broekens, A. Plaat, and C. M. Jonker, "Model-based Reinforcement Learning: A Survey," *Foundations and Trends® in Machine Learning*, 2023. [Online]. Available: https://doi.org/10.1561/2200000086

[18] B. Ai, S. Tian, H. Shi, Y. Wang, T. Pfaff, C. Tan, H. I. Christensen, H. Su, J. Wu, and Y. Li, "A review of learning-based dynamics models for robotic manipulation," *Science Robotics*, 2025. [Online]. Available: https://doi.org/10.1126/scirobotics.adt1497

[19] Y. Gao, Q. Zhang, D. Ding, and D. Zhao, "Dream to Drive With Predictive Individual World Model," *IEEE Transactions on Intelligent Vehicles*, 2024. [Online]. Available: https://doi.org/10.1109/tiv.2024.3408830

[20] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning," *Annual Review of Control Robotics and Autonomous Systems*, 2022. [Online]. Available: https://doi.org/10.1146/annurev-control-042920-020211

[21] G. Dulac-Arnold, N. Levine, D. J. Mankowitz, J. Li, C. Păduraru, S. Gowal, and T. Hester, "Challenges of real-world reinforcement learning: definitions, benchmarks and analysis," *Machine Learning*, 2021. [Online]. Available: https://doi.org/10.1007/s10994-021-05961-4

[22] E. Kaiser, J. N. Kutz, and S. L. Brunton, "Sparse identification of nonlinear dynamics for model predictive control in the low-data limit," *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 474, no. 2219, p. 20180335, 2018. [Online]. Available: https://doi.org/10.1098/rspa.2018.0335